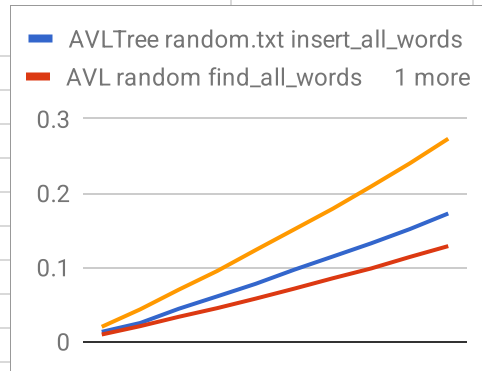
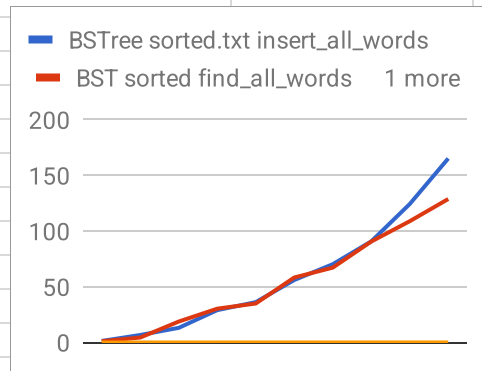
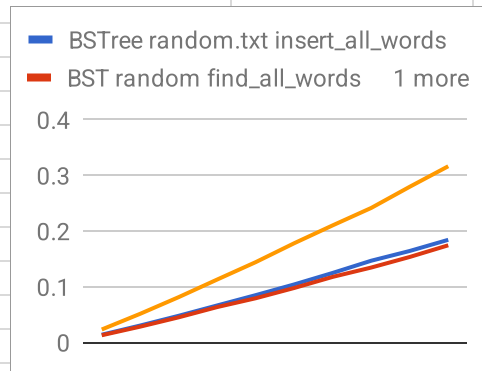
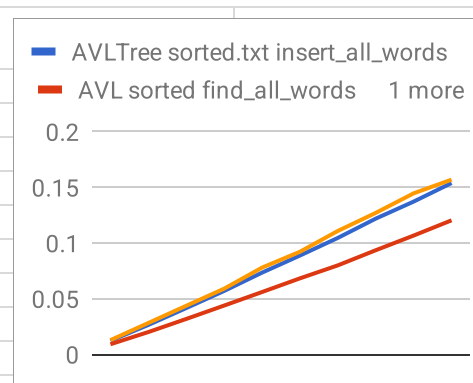


Partition (of N)	BSTree random.txt insert_all_words	BST random find_all_words	BST random remove_all_words	NAME
1	0.01401	0.013014	0.023367	
2	0.030506	0.028492	0.05161	
3	0.04802	0.04525	0.081544	
4	0.066603	0.063671	0.113163	
5	0.085116	0.079495	0.144388	
6	0.104341	0.097939	0.17836	
7	0.125019	0.117901	0.210481	
8	0.146908	0.134664	0.241739	
9	0.164439	0.153658	0.279785	
10	0.184462	0.174623	0.316589	
Partition (of N)	BSTree sorted.txt insert_all_words	BST sorted find_all_words	BST sorted remove_all_words	
1	1.38608	1.14906	0.003554	
2	6.74284	4.45849	0.005968	
3	13.1607	18.8182	0.009087	
4	29.0729	30.3797	0.011841	
5	36.252	35.019	0.014607	
6	56.1123	58.3648	0.017572	
7	70.314	67.2388	0.020409	
8	90.8217	90.6802	0.048089	
9	124.382	108.947	0.051823	
10	165.308	128.876	0.052341	
Partition (of N)	AVLTree random.txt insert_all_words	AVL random find_all_words	AVL random remove_all_words	
1	0.013028	0.009429	0.019828	
2	0.024959	0.020694	0.043342	
3	0.044093	0.033288	0.069819	
4	0.060835	0.044968	0.095039	
5	0.077765	0.057811	0.123292	
6	0.096736	0.071228	0.151042	
7	0.114475	0.085184	0.178877	
8	0.13232	0.098457	0.20901	
9	0.151418	0.113876	0.239991	
10	0.172454	0.128539	0.273227	



Partition (of N)	AVLTree sorted.txt insert_all_words	AVL sorted find_all_words	AVL sorted remove_all_words
1	0.012359	0.00921	0.012896
2	0.026554	0.020161	0.028319
3	0.041705	0.031928	0.043767
4	0.057071	0.043959	0.059113
5	0.073518	0.056107	0.078035
6	0.088882	0.068398	0.092299
7	0.104722	0.080066	0.110984
8	0.122038	0.093658	0.1272
9	0.13733	0.106829	0.144759
10	0.154014	0.12052	0.157063



		Theoretical Time Complexity random.txt	Observed (graphed) Time Complexity random.txt	Theoretical Time Complexity sorted. txt	Observed (graphed) Time Complexity sorted. txt
	class::method()				
	BST::insert()	$O(\log N)$	$O(\log N)$	$O(N)$	$O(N)$
	BST::find()	$O(\log N)$	$O(\log N)$	$O(N)$	$O(N)$
	BST::remove()	$O(\log N)$	$O(\log N)$	$O(1)$	$O(1)$
	AVL::insert()	$O(\log N)$	$O(\log N)$	$O(\log N)$	$O(\log N)$
	AVL::find()	$O(\log N)$	$O(\log N)$	$O(\log N)$	$O(\log N)$
	AVL::remove()	$O(\log N)$	$O(\log N)$	$O(\log N)$	$O(\log N)$